

the number of them which are aszyzygetic; for instance, $i = 5$ as above, $j = 6$, an extract from the table is

j	k	θ	$(k + 1)\theta$
6	30	1	a^6 31
	28	0	0
	26	1	a^4e 27
	24	1	a^2f 25
	22	2	a^3b, a^2bf 46
	20	2	a^4, acf 42
	18	3	a^2d, a^3bc, c^2, f^2 57
	16	2	a^2h, abf, ace 34
	14	4	$a^2b^2, a^2h, acd, bc^2, ef$ 60
	12	3	abe, al, ce, df 39
	10	4	a^2g, abd, b^2c, ch, e^2 44
	8	2	ak, bi, de 18
	6	4	aj, bh, cg, d^2 28
	4	1	n 3
	2	2	bg, m 6
	0	0	0

$$462 = \frac{\Pi(11)}{\Pi(5)\Pi(6)},$$

where, for instance deg-order 6.14, the covariants are $a^2b^2, a^2h, acd, bc^2, ef$, but the number against these in the third column being (not 5 but) 4, the meaning is that there exists between these five terms one syzygy, making the number of aszyzygetic covariants of the deg-order 6.14 to be 4. The second column thus in fact contains the several values of k , and the third column the corresponding values of θ ; whence, forming the several products $(k + 1)\theta$ as shown, the sum of these is as it should be = 462.

Cambridge, 13 July, 1878.

699.

ON THE TRIPLE ϑ -FUNCTIONS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 134–138.]

There should be in all 64 functions proportional to irrational algebraical functions of three independent variables x, y, z ; there is no difficulty in obtaining the expression of these 64 functions in the case of the system of differential equations connected with the integral

$$\int dx : \sqrt{a - x.b - x.c - x.d - x.e - x.f - x.g - x.h - x};$$

but this is *not the general form* of the system for the deficiency (Geschlecht) $p = 3$; and I do not know how to deal with the general form: the present note relates therefore exclusively to the above-mentioned hyper-elliptic form.

I.

If in the Memoir, Weierstrass, “Theorie der Abel’schen Functionen,” Crelle, t. LII. (1856), pp. 285–380, we take $p = 3$, and write $x, y, z; u, v, w; a, b, c, d, e, f, g$ instead of $x_1, x_2, x_3; u_1, u_2, u_3; a_1, a_2, a_3, a_4, a_5, a_6, a_7$; then, neglecting throughout mere constant factors, we have

$$X = a - x.b - x.c - x.d - x.e - x.f - x.g - x,$$

with the like values for Y and Z : the differential equations are

$$\begin{aligned} du &= \frac{b-x.c-x.dx}{\sqrt{X}} + \frac{b-y.c-y.dy}{\sqrt{Y}} + \frac{b-z.c-z.dz}{\sqrt{Z}}, \\ dv &= \frac{c-x.a-x.dx}{\sqrt{X}} + \frac{c-y.a-y.dy}{\sqrt{Y}} + \frac{c-z.a-z.dz}{\sqrt{Z}}, \\ dw &= \frac{a-x.b-x.dx}{\sqrt{X}} + \frac{a-y.b-y.dy}{\sqrt{Y}} + \frac{a-z.b-z.dz}{\sqrt{Z}}. \end{aligned}$$

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and if we write the single letters A, B, C, D, E, F, G for $\text{al}(u, v, w)_1, \text{al}(u, v, w)_2, \text{al}(u, v, w)_3, \text{al}(u, v, w)_4, \text{al}(u, v, w)_5$ respectively, each of the capital letters thus denoting a function of (u, v, w) , the expressions of these functions in terms of (x, y, z) are

$$A = \sqrt{a-x.b-x.c-w}, \quad (\text{seven equations}).$$

Similarly, instead of the 21 functions $\text{al}(u, v, w)_{12}, \dots, \text{al}(u, v, w)_{67}$, writing $AB, \dots, FG$, each of these binary symbols denoting in like manner a function of (u, v, w) , the definition of AB is

$$AB = A \nabla B - B \nabla A,$$

where

$$\nabla = \frac{d}{du} + \frac{d}{dv} + \frac{d}{dw};$$

we have

$$\begin{aligned} \frac{b-c.c-a.a-b}{\sqrt{X}} \nabla x &= \frac{a-y.a-z}{x-y.x-z} (b-c) du + \frac{b-y.b-z}{x-y.x-z} (c-a) dv + \frac{c-y.c-z}{x-y.x-z} (a-b) dw, \\ \frac{dy}{\sqrt{Y}} &= \frac{a-z.a-x}{y-z.y-x} (b-c) du + \frac{b-z.b-x}{y-z.y-x} (c-a) dv + \frac{c-z.c-x}{y-z.y-x} (a-b) dw, \\ \frac{dz}{\sqrt{Z}} &= \frac{a-x.a-y}{z-x.z-y} (b-c) du + \frac{b-x.b-y}{z-x.z-y} (c-a) dv + \frac{c-x.c-y}{z-x.z-y} (a-b) dw; \end{aligned}$$

hence

$$\begin{aligned} \frac{b-c.c-a.a-b}{\sqrt{X}} \nabla x &= \frac{a-y.a-z}{x-y.x-z} (b-c) + \frac{b-y.b-z}{x-y.x-z} (c-a) + \frac{c-y.c-z}{x-y.x-z} (a-b) \\ &= -\frac{b-c.c-a.a-b}{x-y.x-z}, \end{aligned}$$

that is,

$$\nabla x = -\frac{\sqrt{X}}{x-y.x-z},$$

and similarly

$$\nabla y = -\frac{\sqrt{Y}}{y-x.y-z}, \quad \nabla z = -\frac{\sqrt{Z}}{z-x.z-y}.$$

Hence from the equation

$$A = \sqrt{a-x.a-y.a-z}$$

we have

$$\nabla A = -\frac{1}{2}A \left(\frac{1}{a-x} \nabla x + \frac{1}{a-y} \nabla y + \frac{1}{a-z} \nabla z \right),$$

that is,

$$\nabla A = \frac{\frac{1}{2}A}{y-z.z-x.x-y} \left\{ \frac{y-z}{a-x} \sqrt{X} + \frac{z-x}{a-y} \sqrt{Y} + \frac{x-y}{a-z} \sqrt{Z} \right\};$$

and similarly

$$\nabla B = \frac{\frac{1}{2}B}{y-z, z-x, x-y} \left\{ \frac{y-z}{b-x} \sqrt{X} + \frac{z-x}{b-y} \sqrt{Y} + \frac{x-y}{b-z} \sqrt{Z} \right\};$$

consequently

$$AB = \frac{1}{2}(a-b) \frac{AB}{y-z, z-x, x-y} \left\{ \frac{(y-z)\sqrt{X}}{a-x, b-x} + \frac{(z-x)\sqrt{Y}}{a-y, b-y} + \frac{(x-y)\sqrt{Z}}{a-z, b-z} \right\}.$$

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or substituting for A and B their values, and disregarding the constant factor $\frac{1}{2}(a-b)$, this is

$$AB = \frac{1}{y-z, z-x, x-y} \left\{ (y-z) \sqrt{a-y, b-y, a-z, b-z, c-x, d-x, e-x, f-x, g-x} \right. \\ \left. + (z-x) \sqrt{a-z, b-z, a-x, b-x, c-y, d-y, e-y, f-y, g-y} \right. \\ \left. + (x-y) \sqrt{a-x, b-x, a-y, b-y, c-z, d-z, e-z, f-z, g-z} \right\}.$$

We have thus in all 21 equations, which exhibit the form of the Weierstrassian functions $\text{al}(u, v, w)_{12}, \dots, \text{al}(u, v, w)_{67}$.

To complete the system, there should it is clear be 35 new functions $\text{al}(u, v, w)_{123}, \dots, \text{al}(u, v, w)_{567}$, represented by $ABC, \dots, EFG$, viz. the whole number of functions would then be

$$7 + \binom{7}{2} + \binom{7}{3} (= 7 + 21 + 35) = 63, = 64 - 1,$$

since the functions represent ratios of the ϑ -functions.

II.

Starting now with the radical

$$\sqrt{a-x, b-x, c-x, d-x, e-x, f-x, g-x, h-x}$$

composed of eight linear factors, and writing, as in my “Memoir on the double ϑ -functions,” t. LXXXV. (1878), pp. 214–245, [665]; a, b, c, d, e, f, g, h to denote these factors, and similarly $a_1, b_1, c_1, d_1, e_1, f_1, g_1, h_1$ and $a_2, b_2, c_2, d_2, e_2, f_2, g_2, h_2$ to denote $a-y, b-y$, etc., and $a-z, b-z$, etc., so that $X = abcdefgh$, $Y = a_1b_1c_1d_1e_1f_1g_1h_1$, $Z = a_2b_2c_2d_2e_2f_2g_2h_2$; then, instead of the Weierstrassian form, the differential equations may be taken to be

$$du = \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}}, \\ dv = \frac{x dx}{\sqrt{X}} + \frac{y dy}{\sqrt{Y}} + \frac{z dz}{\sqrt{Z}}, \\ dw = \frac{x^2 dx}{\sqrt{X}} + \frac{y^2 dy}{\sqrt{Y}} + \frac{z^2 dz}{\sqrt{Z}}.$$

We then have 64 ϑ -functions and an ω -function, viz. writing

$$\theta = y - z, z - x, x - y,$$

and then

$$\sqrt{a} = \sqrt{aa_1a_2} \quad (8 \text{ equations})$$

$$\sqrt{abc} = \frac{1}{\theta} \{ (y-z)\sqrt{a_1b_1c_1a_2b_2c_2efgh} + (z-x)\sqrt{a_2b_2c_2abcd_1e_1f_1g_1h_1} + (x-y)\sqrt{abc a_1b_1c_1d_2e_2f_2g_2h_2} \}$$

(56 equations)

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the equations, which define the ϑ -functions $A, B, \dots, H, ABC, \dots, FGH$, and the ω -function Ω , are

$$A = \Omega\sqrt{a} \quad (8 \text{ equations})$$

$$ABC = \Omega\sqrt{abc} \quad (56 \text{ equations})$$

and one other relation which I have not as yet investigated.

As regards the algebraical relations between the 64 ϑ -functions, it is to be remarked that, selecting in a proper manner 8 of the functions, the square of any one of the other functions can be expressed as a linear function of the squares of the 8 selected functions. To explain this somewhat further, observe that, taking any 5 squares such as $(ABC)^2$, we can with these 5 squares form a linear combination which is rational in x, y, z . We have for instance, writing down the irrational part only,

$$(ABC)^2 = \frac{2}{\theta^2} \{ abc(z-x)(x-y)\sqrt{YZ} + a_1b_1c_1(y-z)(z-x)\sqrt{ZX} + a_2b_2c_2(x-y)(y-z)\sqrt{XY} \},$$

and forming in all five such equations, then inasmuch as the coefficients $abc, \dots$ of $\sqrt{YZ}(z-x)(x-y)$ are each of them a cubic function containing terms in x^3, x^2, x, x^0 , we have a determinate set of constant factors such that the resulting term in $(z-x)(x-y)\sqrt{YZ}$ will be $= 0$; but the coefficients $a_1b_1c_1, \dots$ of $(x-y)(y-z)\sqrt{ZX}$ only differ from the first set of coefficients by containing y instead of x , and the same set of constant factors will thus make the resulting term in $(x-y)(y-z)\sqrt{ZX}$ to be $= 0$; and similarly the same set of constant factors will make the resulting term in $(y-z)(z-x)\sqrt{XY}$ to be $= 0$; viz. we have thus a set of constant factors, such that the whole irrational part will disappear. It seems to be in general true that the same set of constant factors will make the rational part integral; viz. the rational part is a function of the form $\frac{1}{\theta}$ multiplied by a rational and integral function of x, y, z , and if this rational and integral function divide by θ^2 , then the final result will be a rational and integral function, which, being symmetrical in x, y, z , is at once seen to be a linear function of the symmetrical combinations $1, x+y+z, yz+zx+xy, xyz$. Such a function is obviously a linear function of any four squares A^2, B^2, C^2, D^2 ; or the form is, linear function of five squares $(ABC)^2 = \text{linear function of four squares } A^2$, that is, any one of the five squares is a linear function of 8 squares.

As an instance, consider the three squares $(ABC)^2, (ABD)^2, (ABE)^2$, which are such that we have a linear combination which is rational: in fact, we have here in each function the pair of factors ab , which unites itself with $(z-x)(x-y)\sqrt{XY}$, viz. it is only the coefficient of $ab(z-x)(x-y)\sqrt{XY}$ which has to be made $= 0$: the required combination is obviously

$$(d-e)(ABC)^2 + (e-c)(ABD)^2 + (c-d)(ABE)^2.$$

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Here the irrational part vanishes and the rational part is found to be

$$\begin{aligned} &= \frac{1}{\theta^2} \left[a_1b_1a_2b_2fgh(y-z)^2 \{ (d-e)c_2c_1dc + (e-c)d_2d_1ec + (c-d)e_2e_1dc \} \right. \\ &+ a_2b_2abfgh_1(z-x)^2 \{ (d-e)c_2cd_1e_1 + (e-c)d_2dc_1e_1 + (c-d)e_2ed_1c_1 \} \\ &\left. + aba_1b_1fgh_2(x-y)^2 \{ (d-e)cc_1d_2e_2 + (e-c)dd_1c_2e_2 + (c-d)ee_1d_2c_2 \} \right]. \end{aligned}$$

The three terms in $\{ \}$ are here $= -(c-d)(d-e)(e-c)$ multiplied by $(z-x)(x-y)$, $(x-y)(y-z)$, $(y-z)(z-x)$ respectively; hence the term in $[]$ divides by θ and the result is

$$= -\frac{(c-d)(d-e)(e-c)}{\theta} [a_1 b_1 a_2 b_2 f g h (y-z) + a_2 b_2 a b f_1 g_1 h_1 (z-x) + a b a_1 b_1 f_2 g_2 h_2 (x-y)],$$

or finally this is $= -(c-d)(d-e)(e-c)$

$$\{(a^2 + ab + b^2)fgh - (a^2b + ab^2)(fg + fh + gh) + a^2b^2(f + g + h)\}$$

multiplied by

$$\begin{aligned} &+ (x+y+z)\{-(a+b)fgh + ab(fg + fh + gh) - a^2b^2\} \\ &+ (yz + zx + xy)\{fgh - ab(f + g + h) + a^2b + ab^2\} \\ &+ xyz\{-(fg + fh + gh) + (a+b)(f + g + h) - (a^2 + ab + b^2)\}, \end{aligned}$$

that is, we have $(d-e)(ABC)^2 + (e-c)(ABD)^2 + (c-d)(ABE)^2 =$ a sum of four squares, viz. we have here a linear relation between 7 squares.

I have not as yet investigated the forms of the relations between the products of pairs of ϑ -functions.

Cambridge, 30 September, 1878.

700.

ON THE TETRAHEDROID AS A PARTICULAR CASE OF THE 16-NODAL QUARTIC SURFACE.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 161–164.]

In the paper “Sur un cas particulier de la surface du quatrième ordre avec seize points singuliers,” Crelle, t. LXV. (1866), pp. 284–290, [356], I showed how the surface called the Tetrahedroid could be identified as a special form of Kummer’s 16-nodal quartic surface; but I was not then in possession of the simplified form of the equation of the 16-nodal surface given in my paper “Note sur la surface du quatrième ordre douée de seize points singuliers et de seize plans singuliers,” Crelle, t. LXXIII. (1871), pp. 292, 293, [442]; see also my paper, “A third memoir on Quartic surfaces,” Proc. Lond. Math. Soc. t. III. (1871), p. 250, [454, this Collection, t. VII., p. 281]. Using the equation last referred to, I resume therefore the consideration of the question.

Taking the constants $\alpha, \beta, \gamma, \alpha', \beta', \gamma', \alpha'', \beta'', \gamma''$, such that

$$\alpha + \beta + \gamma = 0, \quad \alpha' + \beta' + \gamma' = 0, \quad \alpha'' + \beta'' + \gamma'' = 0,$$

and writing also

$$\begin{aligned} M &= \alpha\alpha''(\beta' - \gamma') + \beta'\beta''(\gamma' - \alpha') + \gamma'\gamma''(\alpha' - \beta') \\ &= \alpha''\alpha(\beta' - \gamma') + \beta''\beta(\gamma' - \alpha') + \gamma''\gamma(\alpha' - \beta') \\ &= \alpha\alpha'(\beta'' - \gamma'') + \beta\beta'(\gamma'' - \alpha'') + \gamma\gamma'(\alpha'' - \beta'') \\ &= -\frac{1}{2}\{(\beta - \gamma)(\beta' - \gamma')(\beta'' - \gamma'') + (\gamma - \alpha)(\gamma' - \alpha')(\gamma'' - \alpha'') + (\alpha - \beta)(\alpha' - \beta')(\alpha'' - \beta'')\}, \end{aligned}$$

(the equivalence of which different expressions for M is verified without difficulty); writing also X, Y, Z, W as current coordinates, the equation of the 16-nodal surface is

$$0 = \begin{cases} W^2(X^2 + Y^2 + Z^2 - 2YZ - 2ZX - 2XY) \\ + 2W[\alpha\alpha'\alpha''(Y^2Z + YZ^2) + \beta\beta'\beta''(Z^2X + ZX^2) + \gamma\gamma'\gamma''(X^2Y + XY^2) + MXYZ] \\ + (\alpha\alpha'\alpha''YZ + \beta\beta'\beta''ZX + \gamma\gamma'\gamma''XY)^2, \end{cases}$$

where, $\alpha, \beta, \gamma, \alpha', \beta', \gamma', \alpha'', \beta'', \gamma''$ being connected as above, the number of constants is = 6.

The equations of the 16 singular planes are

$$\begin{aligned} X &= 0, & Y &= 0, \\ \alpha(\gamma'\gamma''Y - \beta'\beta''Z) - W &= 0, & \beta(\alpha'\alpha''Z - \gamma'\gamma''X) - W &= 0, \\ \alpha'(\gamma''\gamma Y - \beta''\beta Z) - W &= 0, & \beta'(\alpha''\alpha Z - \gamma''\gamma X) - W &= 0, \\ \alpha''(\gamma\gamma'Y - \beta\beta'Z) - W &= 0, & \beta''(\alpha\alpha'Z - \gamma\gamma'X) - W &= 0, \\ Z &= 0, & W &= 0, \\ \gamma(\beta'\beta''X - \alpha'\alpha''Y) - W &= 0, & \beta\gamma X + \gamma\alpha Y + \alpha\beta Z &= 0, \\ \gamma'(\beta''\beta X - \alpha''\alpha Y) - W &= 0, & \beta'\gamma'X + \gamma'\alpha'Y + \alpha'\beta'Z &= 0, \\ \gamma''(\beta\beta'X - \alpha\alpha'Y) - W &= 0, & \beta''\gamma''X + \gamma''\alpha''Y + \alpha''\beta''Z &= 0. \end{aligned}$$

Writing x, y, z, w as current coordinates, the equation of the Tetrahedroid is

$$0 = \begin{cases} w^4(l^2y^2 + m^2z^2) + l^2m^2n^4w^4 + l^4m^2n^2x^2y^2z^2 + f^2g^2h^2w^4 \\ + (l^2f^2 - m^2g^2 + n^2h^2)(m^2z^2x^2 + g^2y^2w^2) \\ + (-l^2f^2 + m^2g^2 - n^2h^2)(l^2y^2z^2 + f^2x^2w^2) \\ + (-l^2f^2 - m^2g^2 + n^2h^2)(n^2x^2y^2 + h^2z^2w^2) = 0, \end{cases}$$

where, inasmuch as f, g, h, l, m, n enter homogeneously, the number of constants is = 5.

The equations of the 16 singular planes, written in an order corresponding to that used for the 16-nodal surface, are

$$\begin{aligned} ny - mz + fw &= 0, & -nx + lz + gw &= 0, & mx - ly + hw &= 0, & -fx - gy - hz &= 0, \\ fx - gy - hz &= 0, & nx + ly + hw &= 0, & -nx - lz + gw &= 0, & ny - mz - fw &= 0, \\ -mx - ly + hw &= 0, & -fx - gy - hz &= 0, & ny + mz + fw &= 0, & -nx + lz - gw &= 0, \\ nx + lz + gw &= 0, & -ny - mz + fw &= 0, & -fx - gy + hz &= 0, & mx - ly - hw &= 0. \end{aligned}$$

These equations can be made to agree each to each with those of the 16 singular planes of the 16-nodal surface, provided that we have

$$\frac{m}{\gamma} = \frac{n}{\beta}, \quad \frac{n}{\alpha'} = \frac{l}{\gamma'}, \quad \frac{l}{\beta''} = \frac{m}{\alpha''}; \quad f = -l\alpha\alpha'\alpha'', \quad g = -m\beta\beta'\beta'', \quad h = -n\gamma\gamma'\gamma'',$$

where observe that the first three equations give $\alpha\beta'\gamma'' = \alpha''\beta\gamma'$, which is the relation between the constants when the 16-nodal surface reduces itself to a tetrahedroid in the above manner. And if we then assume

$$X = ny - mz + fw, \quad Y = -nx + lz + gw, \quad Z = mx - ly + hw, \quad W = -fx - gy - hz,$$

the 16 linear functions of X, Y, Z, W will become mere constant multiples of the corresponding 16 linear functions of x, y, z, w ; the constants, by which the several

functions of x, y, z, w have to be multiplied in order to reduce them each to the corresponding linear function of X, Y, Z, W , being given by the table

1,	1,	1,	1,
$\frac{1}{m\beta}(l\alpha - m\beta),$	$-\frac{\alpha\alpha''}{m}(l\alpha - m\beta),$	$\frac{\alpha\alpha''}{n}(l\alpha - m\beta),$	$\frac{\beta\gamma}{\alpha}\alpha(\beta'\gamma'' - \gamma'\beta''),$
$\frac{1}{n\gamma}(m\beta - n\gamma),$	$-\frac{\beta\beta''}{n\gamma}(m\beta - n\gamma),$	$\frac{\beta\beta''}{l}(m\beta - n\gamma),$	$\frac{\gamma\alpha}{\beta}(\gamma''\alpha - \gamma\alpha''),$
$\frac{1}{l\alpha}(n\gamma - l\alpha),$	$\frac{\gamma\gamma''}{l\alpha}(n\gamma - l\alpha),$	$-\frac{\gamma\gamma''}{m}(n\gamma - l\alpha),$	$\frac{\alpha\beta}{\gamma}(\alpha\beta' - \alpha'\beta'').$

For instance, we have

$$\alpha(\gamma'\gamma''Y - \beta'\beta''Z) - W = 2\frac{1}{m\beta}(l\alpha - m\beta)(fx - gy - hz),$$

viz. substituting for Y, Z, W their values, the relation is

$$\left\{ \begin{array}{l} m\alpha\beta\gamma'\gamma''(-nx + lz + gw) \\ - m\alpha\beta\beta'\beta''(mx - ly + hw) \\ + m\beta(-fx - gy - hz) \end{array} \right\} = 2(l\alpha - m\beta)(fx - gy - hz).$$

As regards the terms in y, z , and w , the identity is at once verified. As regards the term in x , we should have

$$m\alpha\beta(-n\gamma'\gamma'' - m\beta'\beta'') - (l\alpha - 2m\beta)f = 0,$$

viz. substituting for f its value, $-l\alpha\alpha'\alpha'' = -m\alpha\alpha'\beta''$, the equation divides by $m\alpha$ and we then have

$$(-n\gamma'\gamma'' - m\beta'\beta'') + \alpha'\beta''(l\alpha - 2m\beta) = 0,$$

that is,

$$l\alpha\alpha'\beta'' - m\beta'\beta''(\beta' + 2\alpha') - n\beta\gamma'\gamma'' = 0,$$

or writing herein $m\beta'' = l\alpha''$, $n\gamma' = l\alpha'$, and $\beta' + 2\alpha' = \alpha' - \gamma'$, the equation becomes $\alpha\alpha'\beta'' - \alpha'\alpha\beta'' - \alpha''\beta(\alpha' - \gamma') - \alpha'\beta\gamma'' = 0$, that is, $\alpha'(\alpha\beta'' - \alpha''\beta) = -\alpha'\beta\gamma''$; or writing herein $\alpha''\beta\gamma' = \alpha'\beta'\gamma$, the equation divided by α' becomes $\alpha\beta'' - \alpha''\beta = \beta\gamma'' - \beta''\gamma$, which is true in virtue of $\alpha + \beta + \gamma = 0$ and $\alpha'' + \beta'' + \gamma'' = 0$. And in like manner the several other identities may be verified.

The equation $\alpha'\beta''\gamma = \alpha''\beta\gamma'$ might have been obtained as the condition of the intersection, in a common point, of four of the singular planes of the 16-nodal surface; and when this equation is satisfied, there are in fact four systems each of four planes, such that the four planes of a system meet in a common point: viz. we have

Planes

$$\begin{aligned} X = 0, \quad & \beta\gamma X + \gamma\alpha Y + \alpha\beta Z = 0, \quad \gamma'(\beta''\beta X - \alpha''\alpha Y) - W = 0, \quad \beta''(\alpha\alpha'Z - \gamma\gamma'X) - W = 0, \\ Y = 0, \quad & \gamma(\beta'\beta''X - \alpha'\alpha''Y) - W = 0, \quad \beta'\gamma'X + \gamma'\alpha'Y + \alpha'\beta'Z = 0, \quad \alpha''(\gamma\gamma'Y - \beta\beta'Z) - W = 0, \\ Z = 0, \quad & \beta'(\alpha''\alpha Z - \gamma''\gamma X) - W = 0, \quad \alpha'(\beta'\gamma Y - \beta''\beta Z) - W = 0, \quad \beta''\gamma''X + \gamma''\alpha''Y + \alpha''\beta''Z = 0, \\ W = 0, \quad & \alpha(\gamma'\gamma''Y - \beta'\beta''Z) - W = 0, \quad \beta(\alpha''\alpha Z - \gamma''\gamma X) - W = 0, \quad \gamma''(\beta\beta'X - \alpha\alpha'Y) - W = 0, \end{aligned}$$

$$\begin{array}{cccc} 0, & -\beta, & \gamma, & \alpha'\beta'\gamma'\cdot\alpha, \\ \alpha', & 0, & -\gamma', & \alpha'\beta'\gamma'\cdot\beta', \\ -\alpha'', & \beta'', & 0, & \alpha''\beta''\gamma''\cdot\gamma'', \\ \beta''\cdot\alpha\alpha'\alpha'', & \alpha''\cdot\beta\beta'\beta'', & \gamma''\cdot\gamma\gamma'\gamma'', & 0, \end{array}$$

the four points being in fact the vertices of the tetrahedron formed by the four planes of the tetrahedroid. Observe that, if the singular planes of the 16-nodal surface in their original order are

$$\begin{array}{c} 1, 2, 3, 4, \\ 5, 6, 7, 8, \\ 9, 10, 11, 12, \\ 13, 14, 15, 16, \end{array}$$

then the planes forming the last-mentioned four systems of planes are

$$\begin{array}{c} (1, 8, 11, 14), \\ (2, 7, 12, 13), \\ (3, 6, 9, 16), \\ (4, 5, 10, 15), \end{array}$$

viz. they correspond each of them to a term which in the determinant formed with the 16 symbols would have the sign +.

The equation $\alpha'\beta''\gamma = \alpha''\beta\gamma'$ is evidently not unique. The triads $(\alpha, \beta, \gamma), (\alpha', \beta', \gamma'), (\alpha'', \beta'', \gamma'')$ enter symmetrically into the equation of the 16-nodal surface; by taking the singular planes of one of the surfaces in a different order, the equation would present itself under one or other of the different forms

$$\begin{array}{l} \alpha'\beta''\gamma = \alpha\beta'\gamma'', \quad \alpha'\beta\gamma'' = \alpha\beta''\gamma', \quad \alpha\beta\gamma'' = \alpha'\beta''\gamma, \\ \alpha'\beta\gamma'' = \alpha''\beta'\gamma, \quad \alpha\beta''\gamma' = \alpha'\beta\gamma'', \\ \alpha''\beta'\gamma = \alpha\beta'\gamma''. \end{array}$$

Cambridge, 9 December, 1878.

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ALGORITHM FOR THE CHARACTERISTICS OF THE TRIPLE ϑ -FUNCTIONS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 165–169.]

The characteristics of the triple ϑ -functions may be represented, the 28 odd characteristics by the binary symbols or duads, 12, . . . , 78, and the even ones (other than $\frac{000}{000} = 0$), say the 35 even characteristics, by the ternary symbols or triads 123 . . . 567: which triads may be regarded as abbreviations for the double tetrads 1238.4567, . . . , 5678.1234, the 8 being always attached to the expressed triad. The correspondence of the symbols is given by the diagram

upper line of characteristic								
	000	100	010	110	001	101	011	111
000		236	345	137	467	156	124	257
100	237	67	136	12	157	48	256	35
010	245	127	23	68	134	357	15	47
110	126	13	78	145	356	25	46	234
001	567	146	125	247	45	17	38	26
101	147	58	246	34	16	123	27	367
011	135	347	14	57	28	36	167	456
111	346	24	56	235	37	267	457	18

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Or, what is the same thing, it is

<i>upper line of characteristic</i>								
	000	100	010	110	001	101	011	111
12				100				
13		110						
14			011					
15						010		
16					101			
17								111
18					001			
23			010					
24		111						
25							110	
26						001		
27							101	
28					011			
34				101				
35								100
36							011	
37					111			
38						011		
45					001			
46							110	
47								010
48						100		
56			111					
57				011				
58		101						
67		100						
68				010				
78				110				

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	000	100	010	110	001	101	011	111
123						101		
124							000	
125			001					
126	110							
127		010						
134					010			
135	011							
136			100					
137				000				
145				110				
146		001						
147	101							
156						000		
157					100			
167							011	110
234								
235				111				
236		000						
237	100							
245	010							
246			101					
247				001				
256							100	
257								000
267						111		
345			000					
346	111							
347		011						
356					110			
357						010		
367								101
456								011
457							111	
467				000				
567	001							

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by means of which the two-line-characteristic is at once found when the duad or triad is given.

The new algorithm renders unnecessary the Table I. of Weber’s memoir “Theorie der Abel’schen Functionen vom Geschlecht 3” (Berlin, 1876). In fact, the system of six pairs corresponding to an odd characteristic such as 12 is

$$13.23, 14.24, 15.25, 16.26, 17.27, 18.28,$$

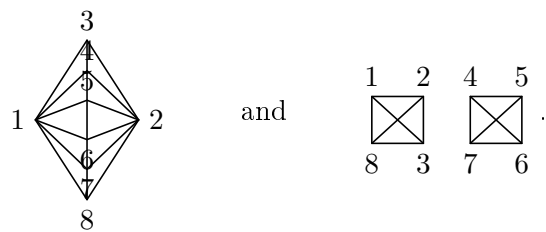
and that corresponding to an even characteristic such as 123 (= 1238.4567) is

$$12.38, 13.28, 18.23, 45.67, 46.57, 47.56;$$

so that all the $(28 + 35 =) 63$ systems can be at once formed.

The odd characteristics correspond to the bitangents of a quartic curve, and as regards these bitangents the notation is, in fact, the notation arising out of Hesse’s investigations and explained Salmon’s *Higher*

Plane Curves (2nd Ed. 1873), pp. 222–225. It may be noticed that the geometrical symbols corresponding to the before-mentioned two systems are:



Hence, selecting out of the first system any two pairs, we have a symbol $\square$: but selecting out of the second system any two pairs, we have a symbol which is either $\square$ or $|||$; so that in each case (Salmon, p. 224) the four bitangents are such that the eight points of contact lie on a conic.

The 28 bitangents of the general quartic curve

$$\sqrt{x_1\xi_1} + \sqrt{x_2\xi_2} + \sqrt{x_3\xi_3} = 0,$$

represented by the equations given by Weber, l.c., pp. 100, 101, and taken in the order in which they are there written down, have for their duad-characteristics

18, 28, 38, 23, 13, 12, 48, 14, 58, 15, 68, 16, 78, 17, 24, 34, 25, 35,

26, 36, 27, 37, 67, 57, 56, 45, 46, 47

respectively. Taking out of any one of the 63 systems three pairs of bitangents at pleasure, these give rise to an equation of the curve of a form such as

$$\sqrt{x_1\xi_1} + \sqrt{x_2\xi_2} + \sqrt{x_3\xi_3} = 0.$$

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and the whole number of the forms of equation is thus = 1260. The triads of pairs which enter into the same equation may be

triads such as 12 . 34, 13 . 42, 14 . 23	1 3 ⊠ 2 4	No. = 70,
„ „ 12 . 34, 13 . 42, 56 . 78	1 2 57 □ 4 3 68	= 630,
„ „ 13 . 23, 14 . 24, 15 . 25	3 1 ⬡ 2 5	= 560,

making the whole number = 1260, as already mentioned.

Cambridge, 7 December, 1878.

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ON THE TRIPLE ϑ -FUNCTIONS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 190–198.]

A QUARTIC curve has the deficiency 3, and depends therefore on the triple ϑ -functions: and these, as functions of 3 arguments, should be connected with functions of 3 points on the curve; but it is easy to understand that it is possible, and may be convenient, to introduce a fourth point, and so regard them as functions of 4 points on the curve: thus in the circle, the functions $\cos u, \sin u$ may be regarded as functions of one point $\cos u = x, \sin u = y$, or as functions of two points,

$$\cos u = xx_1 + yy_1, \quad \sin u = xy_1 - x_1y.$$

And accordingly in Weber's memoir "Theorie der Abel'schen Functionen vom Geschlecht 3," (1876), see p. 156, the triple ϑ -functions are regarded as functions of 4 points on the curve: viz. it is in effect shown that (disregarding constant factors) each of the 64 functions is proportional to a determinant, the four lines of which are algebraical functions of the coordinates of the four points respectively: the form of this determinant being different according as the characteristic of the ϑ -function is odd or even, or say according as the ϑ -function is odd or even. But the geometrical signification of these formulae requires to be developed.

A quartic curve may be touched in six points by a cubic curve: but (Hesse, 1855*) there are two kinds of such tangent cubics, according as the six points of contact are on a conic, or are not on a conic; say we have a conic hexad of points on the quartic, and a cubic hexad of points on the quartic. In either case, three points of the hexad may be assumed at pleasure; we can then in 28 different ways determine the remaining three points of the conic hexad, and in 36 different

* See the two memoirs "Ueber Determinanten und ihre Anwendung in der Geometrie" and "Ueber die Doppeltangenten der Curven vierter Ordnung," Crelle, t. XLIX. (1855).

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ways the remaining three points of the cubic hexad: or what is the same thing, there are 28 systems of cubics touching in a conic hexad, and 36 systems of cubics touching in a cubic hexad. The condition in order that four points of the quartic curve may belong to a hexad (conic or cubic) is given by an equation $\Omega = 0$, where Ω is a determinant the four lines of which are algebraical functions of the coordinates of the four points respectively: but the form of such determinant is different according as the condition belongs to a conic hexad, or to a cubic hexad: we have thus 28 conic determinants and 36 cubic determinants, Ω ; and the 64 ϑ -functions are proportional to constant multiples of these determinants; viz. the odd functions correspond to the conic determinants, and the even functions to the cubic determinants.

First, as to the conic hexads: the points of a conic hexad lie in a conic with the two points of contact of some one of the bitangents of the quartic curve: so that, given any three points of the hexad, these together with the two points of contact of the bitangent determine a conic which meets the quartic in the remaining three points of the hexad. Suppose that a, b, c, f, g, h are linear functions of the coordinates such that the equation of the quartic curve is

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0;$$

then $a = 0, b = 0, c = 0, f = 0, g = 0, h = 0$ are six of the bitangents of the curve, and the bitangent $a = 0$ touches the curve at the two points of intersection of this line with the conic $bg - ch = 0$. The general equation of a conic through these two points $a = 0, bg - ch = 0$, may be written

$$bg - ch + a(Ax + By + Cz) = 0,$$

where for x, y, z we may if we please substitute any three of the six linear functions a, b, c, f, g, h , or any other linear functions of the coordinates (x, y, z) : and the equation may also be written

$$\sqrt{af}(\sqrt{bg} - \sqrt{ch}) + \sqrt{a}(Ax + By + Cz) = 0.$$

Adopting this latter form, and considering the intersections of the conic with the quartic, that is, considering the relation

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0$$

as holding good, we have

$$\begin{aligned} af + bg - ch &= -2\sqrt{afbg}, \\ af - bg + ch &= -2\sqrt{afch}, \\ -af + bg + ch &= -2\sqrt{bgch}, \end{aligned}$$

and we thus have at pleasure one or other of the two equations

$$-2\sqrt{afbg} + \sqrt{a}(Ax + By + Cz) = 0,$$

that is,

$$-2\sqrt{afch} + \sqrt{a}(Ax + By + Cz) = 0,$$

$$-2\sqrt{bgch} + \sqrt{a}(Ax + By + Cz) = 0.$$

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Hence the condition in order that the four points (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) , (x_4, y_4, z_4) , assumed to be points of the quartic, may belong to the conic hexad, may be written

$$\begin{vmatrix} \sqrt{f_1 b_1 g_1}, & x_1 \sqrt{a_1}, & y_1 \sqrt{a_1}, & z_1 \sqrt{a_1} \\ \sqrt{f_2 b_2 g_2}, & x_2 \sqrt{a_2}, & y_2 \sqrt{a_2}, & z_2 \sqrt{a_2} \\ \sqrt{f_3 b_3 g_3}, & x_3 \sqrt{a_3}, & y_3 \sqrt{a_3}, & z_3 \sqrt{a_3} \\ \sqrt{f_4 b_4 g_4}, & x_4 \sqrt{a_4}, & y_4 \sqrt{a_4}, & z_4 \sqrt{a_4} \end{vmatrix} = 0, \quad \text{or} \quad \begin{vmatrix} \sqrt{f_1 c_1 h_1}, & x_1 \sqrt{a_1}, & y_1 \sqrt{a_1}, & z_1 \sqrt{a_1} \\ \sqrt{f_2 c_2 h_2}, & x_2 \sqrt{a_2}, & y_2 \sqrt{a_2}, & z_2 \sqrt{a_2} \\ \sqrt{f_3 c_3 h_3}, & x_3 \sqrt{a_3}, & y_3 \sqrt{a_3}, & z_3 \sqrt{a_3} \\ \sqrt{f_4 c_4 h_4}, & x_4 \sqrt{a_4}, & y_4 \sqrt{a_4}, & z_4 \sqrt{a_4} \end{vmatrix} = 0,$$

where, as before, the x, y, z may be replaced by any three of the letters a, b, c, f, g, h , or by any other linear functions of (x, y, z) : and, moreover, although in obtaining the condition we have used for the quartic the equation

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0,$$

depending upon six bitangents, yet from the process itself it is clear that the condition can only depend upon the particular bitangent $a = 0$: calling the condition $\Omega = 0$, all the forms of condition which belong to the same bitangent $a = 0$, will be essentially identical, that is, the several determinants Ω will differ only by constant factors; or disregarding these constant factors, we have for the bitangent $a = 0$, a single determinant Ω , which may be taken to be any one of the determinants in question. And we have thus 28 determinants Ω , corresponding to the 28 bitangents respectively.

Coming now to the cubic hexads, Hesse showed that the equation of a quartic curve could be (and that in 36 different ways) expressed in the form, symmetrical determinant $= 0$, or say

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix} = 0,$$

where $(a, b, c, d, f, g, h, l, m, n)$ are linear functions of the coordinates; and from each of these forms he obtains the equation of a cubic

$$\begin{vmatrix} a, & h, & g, & l, & \alpha \\ h, & b, & f, & m, & \beta \\ g, & f, & c, & n, & \gamma \\ l, & m, & n, & d, & \delta \\ \alpha, & \beta, & \gamma, & \delta, & \cdot \end{vmatrix} = 0,$$

containing the four constants $\alpha, \beta, \gamma, \delta$, or say the 3 ratios of these constants, touching the quartic in a cubic hexad of points: that the cubic does touch the quartic in six points appears, in fact, from Hesse's identity

$$\begin{vmatrix} a, & h, & g, & l, & \alpha \\ h, & b, & f, & m, & \beta \\ g, & f, & c, & n, & \gamma \\ l, & m, & n, & d, & \delta \\ \alpha, & \beta, & \gamma, & \delta, & \cdot \end{vmatrix} \begin{vmatrix} a, & h, & g, & l, & \alpha' \\ h, & b, & f, & m, & \beta' \\ g, & f, & c, & n, & \gamma' \\ l, & m, & n, & d, & \delta' \\ \alpha', & \beta', & \gamma', & \delta', & \cdot \end{vmatrix} - \begin{vmatrix} a, & h, & g, & l, & \alpha \\ h, & b, & f, & m, & \beta \\ g, & f, & c, & n, & \gamma \\ l, & m, & n, & d, & \delta \\ \alpha', & \beta', & \gamma', & \delta', & \cdot \end{vmatrix}^2 = U \begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix}$$

where U is an easily calculated function of the second order in $a, b, c, d, f, g, h, l, m, n$, and also of the second order in the determinants $\alpha\beta' - \alpha'\beta$, etc.

We can obtain such a form of the equation of the quartic, from the before-mentioned equation

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0;$$

viz. this equation gives

$$\begin{vmatrix} *, & h, & g, & a \\ h, & *, & f, & b \\ g, & f, & *, & c \\ a, & b, & c, & * \end{vmatrix} = 0,$$

which is of the required form, symmetrical determinant = 0; the equation is, in fact,

$$a^2f^2 + b^2g^2 + c^2h^2 - 2bcgh - 2cahf - 2abfg = 0,$$

which is the rationalised form of

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0,$$

and we hence have the cubic

$$\begin{vmatrix} *, & h, & g, & a, & \alpha \\ h, & *, & f, & b, & \beta \\ g, & f, & *, & c, & \gamma \\ a, & b, & c, & *, & \delta \\ \alpha, & \beta, & \gamma, & \delta, & * \end{vmatrix} = 0,$$

the developed form of which is

$$\begin{aligned} & \alpha^2bcf + \beta^2cag + \gamma^2abh + \delta^2fgh \\ & - (\alpha\beta\gamma + f\alpha\delta)(-af + bg + ch) \\ & - (\beta\gamma\alpha + g\beta\delta)(af - bg + ch) \\ & - (\gamma\alpha\beta + h\gamma\delta)(af + bg - ch) = 0. \end{aligned}$$

Considering the intersections with the quartic

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0,$$

we have

$$-af + bg + ch, \quad af - bg + ch, \quad af + bg - ch = -2\sqrt{bcgh}, \quad -2\sqrt{cahf}, \quad -2\sqrt{abfg},$$

and the equation thus becomes

$$(\alpha\sqrt{bcf} + \beta\sqrt{cag} + \gamma\sqrt{abh} + \delta\sqrt{fgh})^2 = 0;$$

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viz. for the points of the cubic hexad we have

$$\alpha\sqrt{bcf} + \beta\sqrt{cag} + \gamma\sqrt{abh} + \delta\sqrt{fgh} = 0,$$

and hence the condition in order that the four points (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) , (x_4, y_4, z_4) may belong to the cubic hexad is

$$\begin{vmatrix} \sqrt{b_1 c_1 f_1}, & \sqrt{c_1 a_1 g_1}, & \sqrt{a_1 b_1 h_1}, & \sqrt{f_1 g_1 h_1} \\ \sqrt{b_2 c_2 f_2}, & \sqrt{c_2 a_2 g_2}, & \sqrt{a_2 b_2 h_2}, & \sqrt{f_2 g_2 h_2} \\ \sqrt{b_3 c_3 f_3}, & \sqrt{c_3 a_3 g_3}, & \sqrt{a_3 b_3 h_3}, & \sqrt{f_3 g_3 h_3} \\ \sqrt{b_4 c_4 f_4}, & \sqrt{c_4 a_4 g_4}, & \sqrt{a_4 b_4 h_4}, & \sqrt{f_4 g_4 h_4} \end{vmatrix} = 0,$$

viz. we have thus the form of the determinant Ω which belongs to a cubic hexad.

It is to be observed that the equation

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0$$

remains unaltered by any of the interchanges a and f , b and g , c and h ; but we thus obtain only two cubic hexads; those answering to the equations

$$\alpha\sqrt{bcf} + \beta\sqrt{cag} + \gamma\sqrt{abh} + \delta\sqrt{fgh} = 0,$$

and

$$\alpha\sqrt{agh} + \beta\sqrt{bhf} + \gamma\sqrt{cfg} + \delta\sqrt{abc} = 0,$$

which give distinct hexads. The whole number of ways in which the equation of the quartic can be expressed in a form such as

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0,$$

attending only to the pairs of bitangents, and disregarding the interchanges of the two bitangents of a pair, is = 1260, and hence the number of forms for the determinant Ω of a cubic hexad is the double of this, = 2520, which is = 36×70 : but the number of distinct hexads is = 36, and thus there must be for each hexad, 70 equivalent forms.

To explain this, observe that every even characteristic except $\frac{000}{000}$, and every odd characteristic, can be (and that in 6 ways) expressed as a sum of two different odd characteristics; we have thus (see Weber's Table I.) a system of $(35 + 28 =) 63$ hexpairs; and selecting at pleasure any three pairs out of the same hexpair, we have a system of $(63 \times 20 =) 1260$ tripairs; giving the 1260 representations of the quartic in a form such as

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0.$$

Each even characteristic (not excluding $\frac{000}{000}$) can be in 56 different ways (Weber, p. 23) expressed as a sum of three different odd characteristics, and these are such that no two of them belong to the same pair, in any tripair; or we may say that each even characteristic gives rise to 56 hemi-tripairs. But a hemi-tripair can be in 5 different ways completed into a tripair; and we have thus, belonging to the same

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even characteristic $(56 \times 5 =) 280$ tripairs, which are however 70 tripairs each taken 4 times. A tripair contains in all $(2^3 =) 8$ hemi-tripairs, but these divide themselves into two sets each of 4 hemi-tripairs such that for each hemi-tripair of the first set the three characteristics have a given sum, and for each hemi-tripair of the second set the three characteristics have a different given sum. Hence considering the 70 tripairs corresponding as above to a given even characteristic, in any one of the 70 tripairs, there is a set of 4 hemi-tripairs such that in each of them the sum of the three characteristics is equal to the given even characteristic; and taking the bitangents f, g, h to correspond to any one of these hemi-tripairs, the bitangents which correspond to the other three hemi-tripairs will be b, c, f ; c, a, g and a, b, h respectively; and we thus obtain from any one of these one and the same representation

$$\alpha\sqrt{bcf} + \beta\sqrt{cag} + \gamma\sqrt{abh} + \delta\sqrt{fgh} = 0$$

of the cubic hexad. And the 70 tripairs give thus the 70 representations of the same cubic hexad.

The whole number of hemi-tripairs is $36 \times 56 = 2016$: it may be remarked that there exists a system of 288 heptads, each of 7 odd characteristics such that selecting at pleasure any 3 characteristics out of the heptad, we obtain always a hemi-tripair; we have thus in all $288 \times 35 = 10080$ hemi-tripairs: this is

$= 2016 \times 5$, or we have the 2016 hemi-tripairs each taken 5 times. Weber’s Table II. exhibits 36 out of the 288 heptads.

I recall that in the algorithm derived from Hesse’s theory the bitangents are represented by the duads $12, 13, \dots, 78$ formed with the eight figures $1, 2, 3, 4, 5, 6, 7, 8$; these duads correspond to the odd characteristics as shown in the Table, and the table shows also triads corresponding to all the even characteristics except $\frac{000}{000}$.

Top line of characteristic.

	000	100	010	110	001	101	011	111
000		236	345	137	467	156	124	257
100	237	67	136	12	157	48	256	35
010	245	127	23	68	134	357	15	47
110	126	13	78	145	356	25	46	234
001	567	146	125	247	45	17	38	26
101	147	58	246	34	16	123	27	367
011	135	347	14	57	17	36	167	456
111	346	24	56	235	37	267	457	18

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See my “Algorithm of the triple ϑ -functions,” Crelle, t. LXXXVII. p. 165, [701]. The $(35 + 28 =)$ 63 hexpairs then are

35 hexpairs such as $\begin{smallmatrix} 1 & 2 \\ \boxtimes & \\ 4 & 3 \end{smallmatrix} \quad \begin{smallmatrix} 5 & 6 \\ \boxtimes & \\ 8 & 7 \end{smallmatrix}$, say this is 1234.5678 or for shortness 567 (the 8 going always with the expressed triad): that is, 567 denotes the hexpair

$$12.34; 13.24; 14.23; 56.78; 57.68; 58.67;$$

and

28 hexpairs such as $1 \begin{smallmatrix} & 3 \\ 4 \swarrow & \nearrow 5 \\ \diamond & \\ 7 \swarrow & \nearrow 6 \\ & 8 \end{smallmatrix} 2$, say this is 12; that is, 12 denotes the hexpair

$$13.32; 14.42; 15.52; 16.62; 17.72; 18.82.$$

It is to be noticed that the odd characteristics, as represented by their duad symbols, can be added by the formulae

$$12 + 23 = 13, \text{ etc.,}$$

or, what is the same thing,

$$12 + 13 + 23 = 0, \text{ etc.,} \quad (= \frac{000}{000}),$$

and

$$12 + 34 = 13 + 24 = 14 + 23 = 56 + 78 = 57 + 68 = 58 + 67 = 567, \text{ etc.}$$

Thus, referring to the table,

$$12 + 23 = 13 \quad \text{means} \quad \frac{110}{100} + \frac{010}{010} = \frac{100}{110},$$

and

$$12 + 34 = 567 \quad \text{means} \quad \frac{110}{100} + \frac{110}{101} = \frac{000}{001},$$

which are right.

The 288 heptads are

8 heptads such as $\begin{array}{c} 1 \\ \diagup \diagdown \diagup \diagdown \diagup \diagdown \diagup \diagdown \\ 2345678 \end{array}$, say this is the heptad 1, denoting the seven duads 12, 13, 14, 15, 16, 17, 18:

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and

280 heptads such as $\begin{array}{c} 1 \\ \diagup \diagdown \diagup \diagdown \\ 2345 \end{array}$ $\begin{array}{c} 6 \\ \diagup \diagdown \\ 7 \quad 8 \end{array}$, say this is the heptad 1.678, denoting the seven duads 12, 13, 14, 15, 67, 68, 78.

We hence see that the 2016 hemi-tripairs are

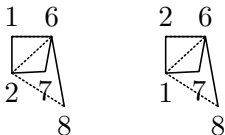
280 hemi-tripairs $\begin{array}{c} 1 \\ \diagup \diagdown \diagup \diagdown \\ 2 \quad 3 \quad 4 \end{array}$ (I.), say this is 1.234, denoting the three duads 12, 13, 14:

1680 hemi-tripairs $\begin{array}{c} 1 \quad 6 \\ / \quad \backslash \\ 2 \quad 7 \quad 8 \end{array}$ (II.), say this is 12(6.78), denoting the three duads 12, 67, 68:

56 hemi-tripairs $\begin{array}{c} 1 \\ \diagup \diagdown \\ 2 \quad 3 \end{array}$ (III.), say this is 123, denoting the three duads 12, 13, 23;

2016.

We further see how each hemi-tripair may be completed into a tripair in 5 different ways: thus (I.) gives the 5 tripairs $2 \cdot \backslash 1 \cdot 4$, $2 \cdot 3 \cdot 4$ (depending on choice 5, 6, 7 or 8); (III.) gives the 5 tripairs $\begin{array}{c} 34 \\ \text{or} \\ (7)8 \end{array}$; while (II.) gives the 3 tripairs $\begin{array}{c} 4a \\ 5 \\ \text{or} \\ 4 \text{ or } 3 \end{array}$ and the 2 tripairs



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To each even characteristic there belongs a system of 56 hemi-tripairs; thus for the characteristic $\frac{000}{000}$, the 56 hemi-tripairs are 123, that is, 12, 13, 23, etc.: whence the 70 tripairs are 1234, that is, 12.34; 13.24; 14.23, etc.; and in any such tripair, say in 1234, we have the set of four hemi-tripairs 123, 124, 134, 234, for each of which the sum of the three characteristics is

$$(12 + 13 + 23 = \frac{000}{000}, \text{ etc.}) :$$

and the other set 1.234, 2.134, 3.124, 4.123, for each of which the sum of the three characteristics is

$$= 567 \ (12 + 13 + 14, = 23 + 14, = 567, = \frac{000}{001}).$$

To find the hemi-tripairs that belong to any other even characteristic; for instance, $\frac{000}{001}$, corresponds to 567: we have 4 such as 1.234; 24 such as (5.12)34; 4 such as 5.678; and 24 such as (1.56)78; in all $4 + 24 + 4 + 24, = 56$. The tripairs are the 2, 1234, 5678; 16 such as 54(123); 16 such as 15(678); 36 such as (5162)34.78; in all $2 + 16 + 16 + 36, = 70$; and in each of these it is easy to select the hemi-tripairs for which the sum of the 3 duads is = 567.

Cambridge, 27 December, 1878.

703.

ON THE ADDITION OF THE DOUBLE ϑ -FUNCTIONS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVIII. (1879), pp. 74–81.]

I ASSUME in general

$$\Theta_e = a - e.b - e.c - e.d - e.e - \theta.f - \theta,$$

and I consider the variables x, y, z, w, p, q , connected by the equations

$$\begin{vmatrix} 1, & 1, & 1, & 1, & 1, & 1 \\ x, & y, & z, & w, & p, & q \\ x^2, & y^2, & z^2, & w^2, & p^2, & q^2 \\ x^3, & y^3, & z^3, & w^3, & p^3, & q^3 \\ \sqrt{X}, & \sqrt{Y}, & \sqrt{Z}, & \sqrt{W}, & \sqrt{P}, & \sqrt{Q} \end{vmatrix} = 0,$$

equivalent to two independent equations, which in fact serve to determine p, q , or say the symmetrical functions $p + q$ and pq , in terms of x, y, z, w .

These equations, it is well known, constitute a particular integral of the differential equations

$$\begin{aligned} \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} + \frac{dw}{\sqrt{W}} + \frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} &= 0, \\ \frac{x dx}{\sqrt{X}} + \frac{y dy}{\sqrt{Y}} + \frac{z dz}{\sqrt{Z}} + \frac{w dw}{\sqrt{W}} + \frac{p dp}{\sqrt{P}} + \frac{q dq}{\sqrt{Q}} &= 0, \end{aligned}$$

or what is the same thing, regarding p, q as arbitrary constants, they constitute the general integral of the differential equations

$$\begin{aligned} \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} + \frac{dw}{\sqrt{W}} &= 0, \\ \frac{x dx}{\sqrt{X}} + \frac{y dy}{\sqrt{Y}} + \frac{z dz}{\sqrt{Z}} + \frac{w dw}{\sqrt{W}} &= 0. \end{aligned}$$